

Automated Test Report

29-12-2025

DUT Name	BBNL MPPT 1kW – [MPPT DC/DC Converter]
DUT Part number	TPS1030004A
DUT Rev & Checksum	S04 – FFABD
Testing person	VISHNU C V

Test Report Summary

SI No	Test Name	Result
1	Input voltage range verification	Fail
2	Output voltage accuracy	Fail
3	Load regulation	Pass



PowerSafe-X

Test Item 1: Input voltage range verification

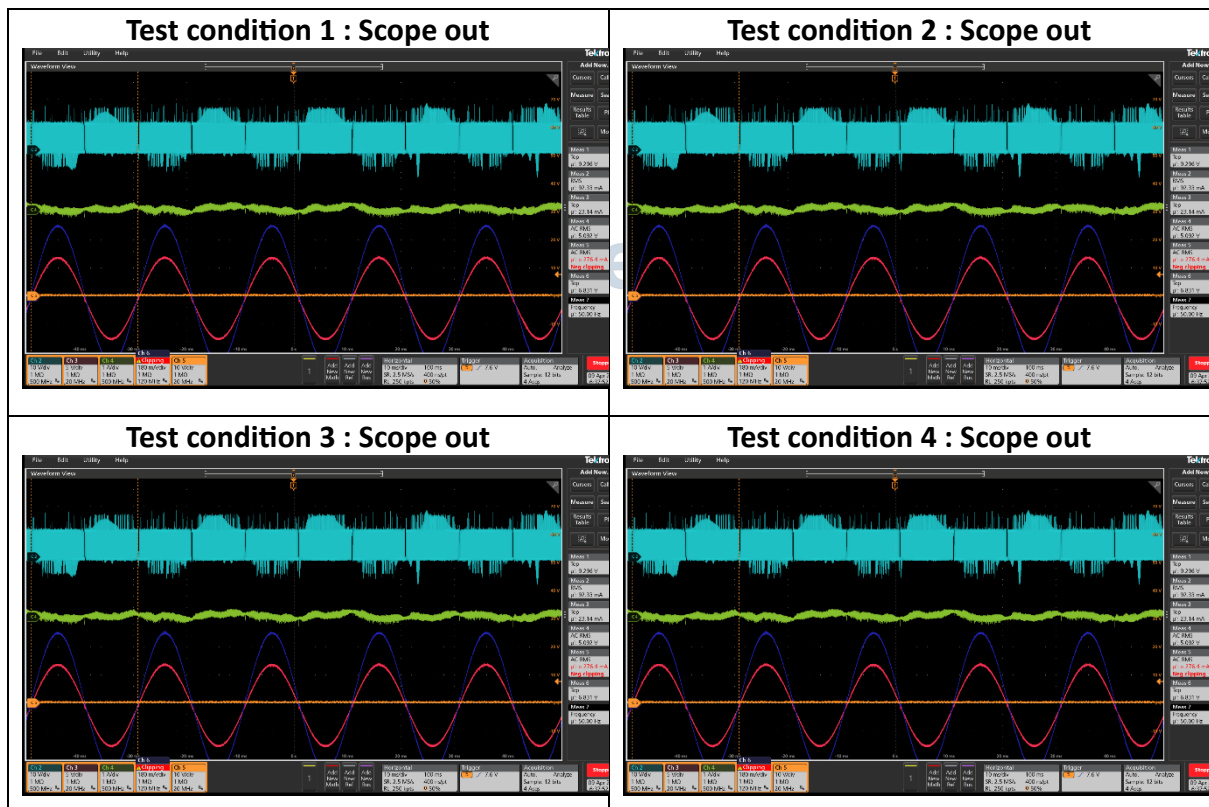
Verification of Input voltage range verification

Test Spec:

- Input voltage =
- Output load =

Test Summary

- Test condition 1 : <<Test condition>>
- Test condition 2 : <<Test condition>>
- Test condition 3 : <<Test condition>>
- Test condition 4 : <<Test condition>>



Timestamp: XXXXXXXX

Test Result: **Fail**

Test Item 2: Output voltage accuracy

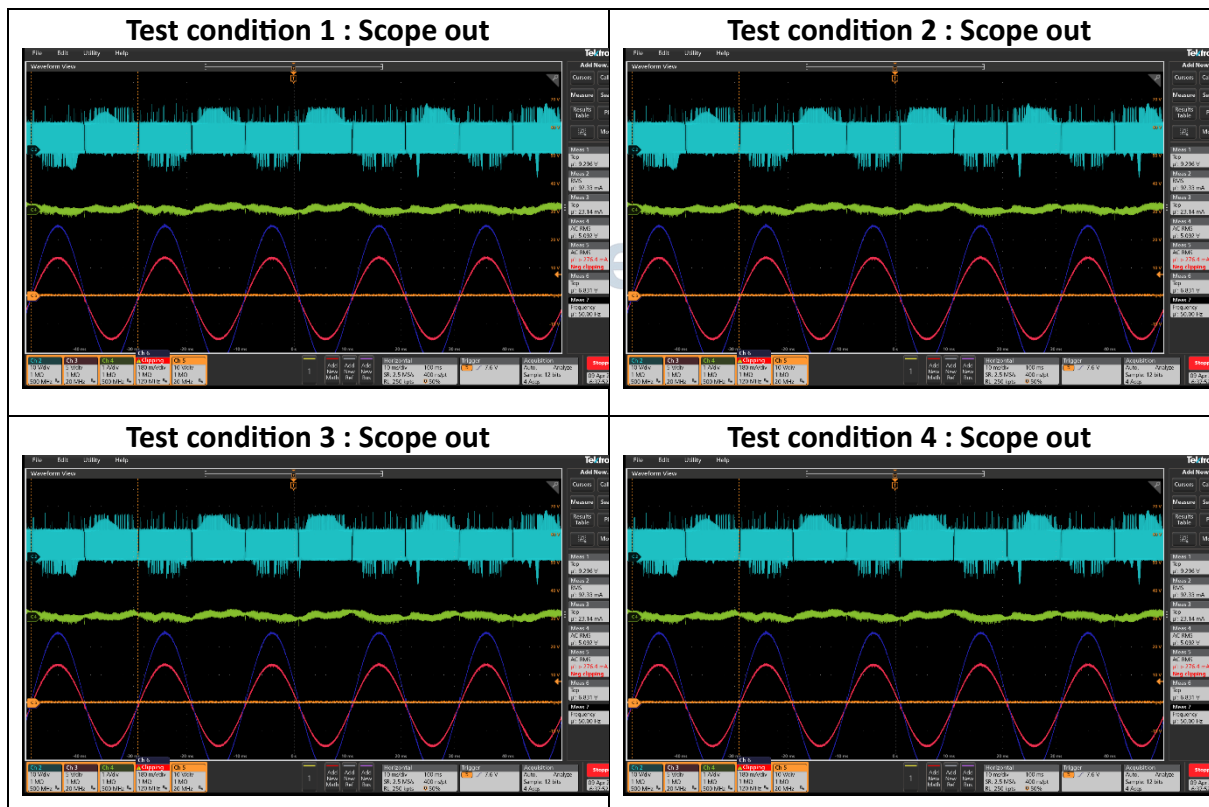
Verification of Input voltage range verification

Test Spec:

- Input voltage =
- Output load =

Test Summary

- Test condition 1 : <<Test condition>>
- Test condition 2 : <<Test condition>>
- Test condition 3 : <<Test condition>>
- Test condition 4 : <<Test condition>>



Timestamp: XXXXXXXX

Test Result: **Fail**

Test Item 3: Load regulation

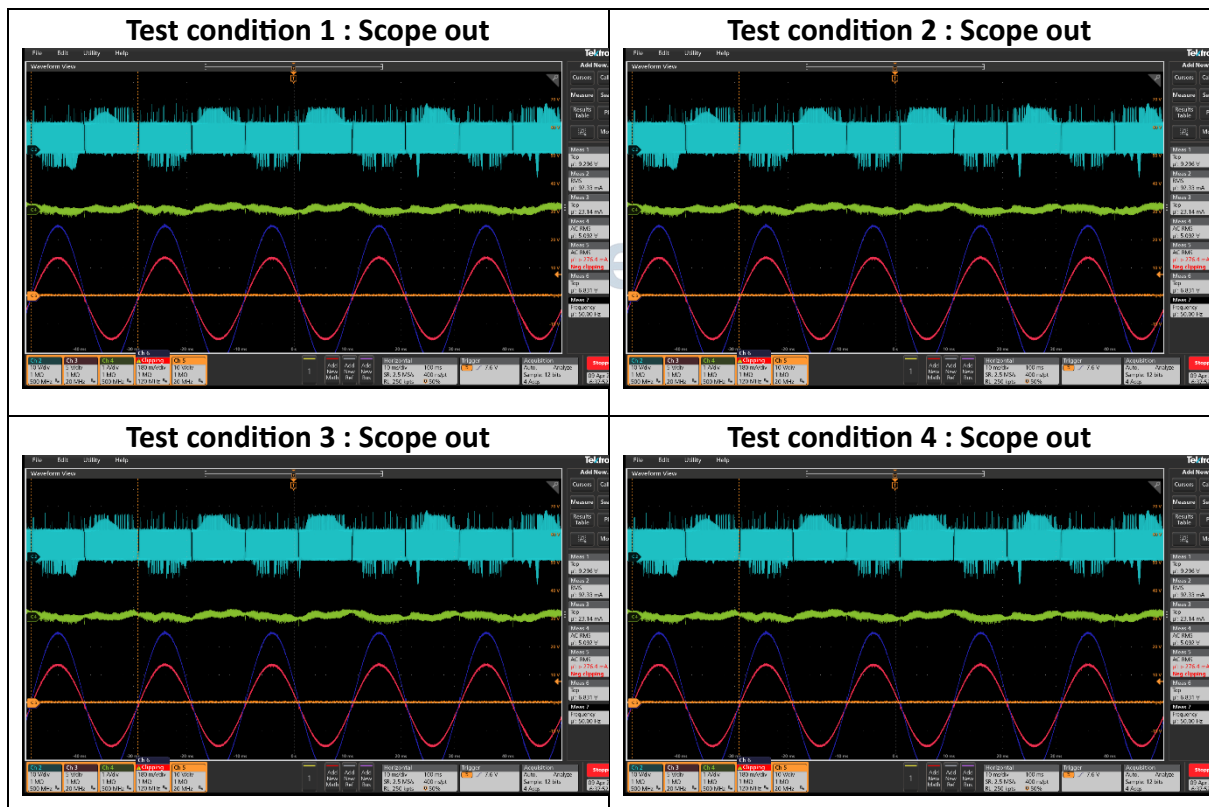
Verification of Input voltage range verification

Test Spec:

- Input voltage =
- Output load =

Test Summary

- Test condition 1 : <<Test condition>>
- Test condition 2 : <<Test condition>>
- Test condition 3 : <<Test condition>>
- Test condition 4 : <<Test condition>>



Timestamp: XXXXXXXX

Test Result: **Pass**

Test log

[16:53:26] Initializing Instruments...
[16:53:26] --- Starting Test: Input voltage range verification ---
[16:53:26] Running Global Setup (130% safety limits)...
[16:53:26] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[16:53:26] Setting Load Limits: I_{max}=23.85A
[16:53:26] Step 1: Setting Source Voltage to Nominal (80.0V)
[16:53:26] Step 2: Load Config -> Mode: CC, Value: 2.0
[16:53:26] Step 3: Turning Source ON
[16:53:26] Step 4: Waiting for V_{out} to reach 95% threshold (51.77V)...
[16:53:36] Timeout waiting for condition after 10s
[16:53:36] Error: Timeout waiting for V_{out} to reach 95% nominal
[16:53:36] Capturing screenshot: D:\Vishnu\Antigravity Workspace\Test Automation GUI\reports\captures\20251229_165336_Input_voltage_range_verification.png
[16:54:08] Screenshot failed: VI_ERROR_TMO (-1073807339): Timeout expired before operation completed.
[16:54:09] --- Starting Test: Output voltage accuracy ---
[16:54:09] Running Global Setup (130% safety limits)...
[16:54:09] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[16:54:09] Setting Load Limits: I_{max}=23.85A
[16:54:09] Step 1: Setting Source Voltage to 48.0V
[16:54:09] Step 2: Turning Source ON
[16:54:10] Step 3: Measuring output voltage at no load
[16:54:10] No Load V_{out}: 23.947V
[16:54:10] Error: 56.06%
[16:54:10] Step 4: Testing at 25% load (4.59A)
[16:54:12] V_{out}: 24.039V, Error: 55.89%
[16:54:12] X FAIL: Exceeds 5.0% tolerance
[16:54:12] Step 5: Testing at 50% load (9.17A)
[16:54:14] V_{out}: 23.908V, Error: 56.13%
[16:54:14] X FAIL: Exceeds 5.0% tolerance
[16:54:14] Step 6: Testing at 75% load (13.76A)
[16:54:16] V_{out}: 24.074V, Error: 55.83%
[16:54:16] X FAIL: Exceeds 5.0% tolerance
[16:54:16] Step 7: Testing at 100% load (18.35A)
[16:54:18] V_{out}: 24.063V, Error: 55.85%
[16:54:18] X FAIL: Exceeds 5.0% tolerance
[16:54:18] X OVERALL FAIL: Output voltage accuracy out of specification
[16:54:18] Capturing screenshot: D:\Vishnu\Antigravity Workspace\Test Automation GUI\reports\captures\20251229_165418_Output_voltage_accuracy.png
[16:54:50] Screenshot failed: VI_ERROR_TMO (-1073807339): Timeout expired before operation completed.
[16:54:51] --- Starting Test: Load regulation ---
[16:54:51] Running Global Setup (130% safety limits)...
[16:54:51] Setting Source Limits: I_{max}=16.25A, P_{max}=1300.00W
[16:54:51] Setting Load Limits: I_{max}=23.85A
[16:54:51] Running Load regulation...
[16:54:53] Capturing screenshot: D:\Vishnu\Antigravity Workspace\Test Automation GUI\reports\captures\20251229_165453_Load_regulation.png
[16:55:25] Screenshot failed: VI_ERROR_TMO (-1073807339): Timeout expired before operation completed.

